

SHAILZA JOLLY

📍 Berlin, Germany

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PROFILE

ML Scientist/Engineer with a Ph.D. in CS and 6+ years spanning academic research and industry. Specializes in LLMs, Generative AI, and scalable data pipelines, with hands-on depth in modern LLM architectures, efficiency techniques (KV-cache, speculative decoding, FlashAttention), and agentic system design. Pairs a track record of top-tier publications (AAAI, NAACL, EMNLP) with shipping production systems and translating technical work into clear business outcomes. I'm exploring what's next at the intersection of research, engineering, and product, open to strong roles and to building something of my own.

SKILLS

- **ML/GenAI:** LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation.
- **LLM systems:** Agentic workflows (MCP, LangGraph/LangSmith), RAG/grounding concepts, experiment design & offline evaluation.
- **Data & scale:** Python, PyTorch, PySpark; large-scale text processing/data quality.
- **Search and retrieval:** embedding models (e.g., Sentence Transformers), reranking, vector search.
- **Platforms:** AWS; production-grade coding, documentation, and cross-functional delivery.

WORK EXPERIENCE

- Applied AI & LLM Systems**
Independent

July 2025 - Present
Berlin, Germany

 - Worked hands-on at the intersection of science, engineering, and product, building and iterating on a series of prototypes across agentic systems, retrieval, and LLM evaluation to keep my engineering sharp while exploring directions worth committing to.
 - Refreshed modern LLM foundations (Stanford) and went deep on architecture and efficiency internals. One example prototype: Sift, an agentic RAG system that returns grounded, citable answers over a corpus of technical PDFs.
- Flinn.ai**
Senior AI Engineer

January 2025 - June 2025
Berlin, Germany

 - Led development of AI-driven product features including data extraction from medical research papers and multilingual complaint monitoring.
 - Partnered with product and backend teams to scope problems, define success metrics, and deliver end-to-end features within an agile product lifecycle.
- Parental Leave (Maternity)**

December 2023 - December 2024

 - Parental leave; stayed engaged with research literature and maintained technical skills.
- Amazon AI**
Research Scientist

July 2022 - November 2023
Berlin, Germany

 - Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
 - Wrote research plans and technical documentation; presented results and recommendations to senior science/engineering leadership.
 - Mentored Master's/Ph.D. interns and collaborated on research deliverables.
- German Research Center for Artificial Intelligence (DFKI)**
Machine Learning Scientist

March 2019 - June 2022
Berlin, Germany

 - Delivered ML research and engineering (coding, experimentation, publications) in BMBF-funded projects: *XAINES* (Explainable AI) and *DeFuseNN* (vision-language systems).
 - Supervised interns and BSc/MSc students; provided technical mentorship and project guidance.

NVIDIA Research
Machine Learning Scientist Intern

May - August 2021
Santa Clara, United States

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

Amazon Alexa
Applied Scientist Intern

August 2019 - December 2019
Aachen, Germany

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

SAP AI Research
Research Intern / Master's Thesis

May 2018 - February 2019
Berlin, Germany

- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

EDUCATION

TU Kaiserslautern, Germany
Ph.D. in Computer Science

March 2019 - June 2022

Building Natural Language Generation and Understanding Systems in Data-Constrained Settings; work on VQA, interpretability, and conversational AI.

Overall grade: **“Sehr Gut” 1.0** (1.0 is highest on 1.0–5.0 scale)

University of Copenhagen, Denmark
Visiting Researcher

January 2021 - March 2021

Developed an unsupervised post-editing algorithm to generate fluent fact-checking explanations.

TU Kaiserslautern, Germany
M.Sc. in Computer Science — Minor in Economics

October 2016 - November 2018

Overall grade: **“Sehr Gut” 1.5**

Kyushu University, Japan
Semester Abroad

November 2017 - February 2018

Explainable AI to analyze behavior of deep CNN architectures for image recognition.

Guru Nanak Dev Engineering College, India
B.Tech in Computer Science & Engineering

August 2012 - July 2016

Overall grade: **First Division with Distinction**

SELECTED PUBLICATIONS

- **Jolly, S.**, Zhang, Z. X., Dengel, A., & Mou, L. (2022). Search and learn: Improving semantic coverage for data-to-text generation. *AAAI*.
- **Jolly, S.**, Pezzelle, S., & Nabi, M. (2021). EaSe: A diagnostic tool for VQA based on answer diversity. *NAACL-HLT*.
- **Jolly, S.**, & Kapoor, S. (2020). Can pre-training help VQA with lexical variations? *Findings of EMNLP*.
- **Jolly, S.**, Falke, T., Tirkaz, C., & Sorokin, D. (2020). Data-efficient paraphrase generation to bootstrap intent classification and slot labeling. *COLING (Industry Track)*.
- **Jolly, S.**, Iwana, B. K., Kuroki, R., & Uchida, S. (2018). How do convolutional neural networks learn design? *ICPR. (Best Student Paper)*

AWARDS AND HONORS

- Received **AAAI-22 Scholarship** by Hitachi to attend AAAI 2022 (2022).
- Awarded **AI Newcomer 2021 Award** (German Informatics Society and BMBF, Germany) (2021).
- Received **EU-Cost STSM Grant** to work on Multi3generation project at CopeNLU (2020).
- **Best Student Paper Award** at ICPR (2018).